Module 3 Lab 2: Service Selection Under Constraints

20 points. Estimated time 90 to 120 minutes.

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How to use this file. Type your answers straight into the shaded boxes. Work down the page in order: each numbered part matches a step in the lab. Do not delete the questions. When you are finished, save this file and upload it as your submission.

1. Your prediction, written first

Step 2 of the lab asks for this before you fill in the table. A wrong guess reasoned honestly earns full marks. A guess edited afterwards to match earns none.

Which factor will appear most often: compliance

How many of the eight (0 to 8): 7

One sentence on why you picked that factor

I assume that is the most costly to companies so it would be prominent in the training materials

2. Your eight decisions

All eight requests are listed below. The latency, volume, and constraint columns are in m3_service_requests.csv. For each one: the obvious answer, your choice, and the factor. The factor must be one of capability-gap, latency, cost, language, compliance, output-shape, spelled exactly.

R1. Read handwritten delivery notes photographed by drivers and pull out the recipient name and the signature date

Decision: Amazon Textract: AnalyzeDocument (FORMS)

Runner up: Amazon Rekognition: DetectText

Why the obvious answer fails. Quote the constraint column

Why the obvious answer fails: DetectText reads text in images but does not reliably output named form-field values and their key-value relationships. Textract FORMS produces structured key-value pairs for recipient name and signature date.

Deciding factor: output-shape

R2. Transcribe recorded pharmacy consultations and flag every medication mentioned

Decision: Transcribe Medical → Comprehend Medical → S3

Runner up: Standard Amazon Transcribe

Why the obvious answer fails. Quote the constraint column

Why the obvious answer fails: Standard transcription alone does not extract medication mentions or address the clinical workflow. Use HIPAA-eligible services with a signed BAA, suitable access controls, and encrypted storage; clinical terminology also favors Transcribe Medical. Deciding factor: compliance

R3. Answer inbound customer phone calls with a voice bot that can book an appointment

Decision:	Amazon Connect → Lex V2 → Lambda → Lex V2	**Runner up:**	Amazon Bedrock general-purpose LLM

Why the obvious answer fails. Quote the constraint column

Why the obvious answer fails: An essay-style LLM response does not by itself handle phone calls, structured slot filling, booking logic, and spoken confirmation. Lex manages intents/slots and voice; Lambda performs booking. Validate the 800 ms requirement end-to-end. Deciding factor: output-shape

R4. Translate a 60,000 word product catalog into six languages every week

Decision:	Amazon Translate (batch) + custom terminology	**Runner up:**	Premium generative translation via Bedrock

Why the obvious answer fails. Quote the constraint column

Why the obvious answer fails: A premium LLM may translate but can be too expensive for 1,440,000 words per month. Amazon Translate is purpose-built for translation; compare character-based pricing to the $300 monthly cap. Deciding factor: cost

R5. Block uploaded profile photos that contain nudity or graphic violence before they appear on the site

Decision:	Rekognition: DetectModerationLabels → policy rules	**Runner up:**	Rekognition: DetectLabels

Why the obvious answer fails. Quote the constraint column

Why the obvious answer fails: DetectLabels describes image contents rather than assigning moderation/safety categories for nudity or graphic violence. DetectModerationLabels returns moderation labels and confidence scores for a block/review policy. Benchmark against 300 ms. Deciding factor: capability-gap

R6. Find personally identifiable information in customer chat transcripts where roughly 40 percent of the chats are in Spanish

Decision:	Comprehend: DetectDominantLanguage → DetectPiiEntities	**Runner up:**	Comprehend: DetectPiiEntities with English assumed

Why the obvious answer fails. Quote the constraint column

> Why the obvious answer fails: An English-only assumption can miss PII in the approximately 40% Spanish transcripts. Identify each transcript's language first, then pass its code to the PII detector.
>
> Deciding factor: language

R7. Show a live human agent a running sentiment read on the chat they are handling so they can escalate

Decision:	Comprehend: synchronous DetectSentiment → agent dashboard	**Runner up:**	Comprehend asynchronous batch sentiment

Why the obvious answer fails. Quote the constraint column

> Why the obvious answer fails: Batch results arrive after processing, not while an agent is handling a live chat. Use synchronous per-message classification and stream updates; benchmark 500 ms, with a real-time SageMaker endpoint as fallback if needed.
>
> Deciding factor: latency

R8. Turn 400 page quarterly compliance filings into a one page brief for the board

Decision:	Textract (if scanned) → Bedrock sections → Bedrock brief	**Runner up:**	Comprehend entity/key-phrase extraction

Why the obvious answer fails. Quote the constraint column

> Why the obvious answer fails: Extracted phrases are not a coherent one-page briefing. Extract the filing text when needed, summarize sections, then synthesize a one-page original-prose board brief with source references.
>
> Deciding factor: output-shape

3. Your chains

Step 3 of the lab. At least two of the eight need more than one service. Name the request, write the chain in order, and say what each stage produces.

Constrained request 1

Request ID:	R2	**The chain, in order:**	Transcribe Medical → Comprehend Medical → S3

Constrained request 2

Request ID: R6

The chain, in order: DetectDominantLanguage → DetectPiiEntities

What each stage of each chain produces, one line per stage

R2 — pharmacy consultations
1. Amazon Transcribe Medical: produces a clinical speech-to-text transcript.
2. Amazon Comprehend Medical: produces structured medication mentions and related details.
3. Amazon S3: stores the transcript and extracted data with appropriate HIPAA safeguards.
R6 — multilingual PII
1. Comprehend DetectDominantLanguage: produces a language code (en or es).
2. Comprehend DetectPiiEntities: produces PII categories, offsets and confidence scores.
Additional chain — R3: appointment voice bot
1. Amazon Connect: receives and routes the incoming phone call.
2. Amazon Lex V2: produces intent and appointment slots from speech.
3. AWS Lambda: produces a booking decision and structured confirmation.
4. Amazon Lex V2 (via Connect): produces the spoken confirmation.
Additional chain — R8: compliance summaries
1. Amazon Textract (when scanned): produces machine-readable filing text.
2. Amazon Bedrock, per section: produces concise section summaries.
3. Amazon Bedrock, final pass: produces a one-page prose board brief.

4. Your console verification

Step 4 of the lab. One service, run twice. Screenshots go in your submission as separate image files or collected into one PDF.

Service I ran: Textract with forms

Confidence on the easy input: 81%

Confidence on the awkward input: 90%

The awkward input I built, written out in full

Its my signature, which is cursive and does not contain all letters fully formed. It returned Hu Sa Ka Nea

What changed between the two results, and whether the service warned me

The awkward input was written in cursive and was not fully formed

5. Your two cost estimates

Step 5 of the lab. The two highest volume requests. Arithmetic line by line, then the source and the date you read it.

Constrained request 1

Request ID:	R7	**Monthly total:**	$1,850 estimated

Constrained request 2

Request ID:	R5	**Monthly total:**	$1,880 estimated

The arithmetic, line by line

R7 — Amazon Comprehend DetectSentiment (9 million messages)

Assume each message is 300 characters or less, so each call is billed for at least 3 units (100 characters per unit).

9,000,000 messages × 3 units = 27,000,000 units

First 10,000,000 × $0.0001 = $1,000

Next 17,000,000 × $0.00005 = $850

Total = $1,850/month

R5 — Amazon Rekognition DetectModerationLabels (2.1 million photos)

First 1,000,000 × $0.001 = $1,000

Next 1,100,000 × $0.0008 = $880

Total = $1,880/month

US East

Pricing page URL and the date I read it

R7: https://aws.amazon.com/comprehend/pricing/
R5: https://aws.amazon.com/rekognition/pricing/
11/7/26

One sentence: is this plausible for that business

R7: For a big support team, around $1,850 a month seems reasonable if it helps agents catch problems early.
R5: Around $1,880 seems reasonable for a platform checking millions of photos, compared with reviewing them all by hand.

6. Count what actually decided it

Step 2 again, at the end. Count how many times each factor appeared in your finished table, then compare against the prediction you wrote at the top of this file.

The factor that appeared most often: output-shape

How many times, out of eight: 3 of 8

7. Break it on purpose

Working code shows you what happens. Broken code shows you why. Do each of these deliberately, paste the real error text, and undo the change before moving on.

Break 1. Take the service you verified in Step 4 and send it an empty input, then an input in a language it does not support.

Paste the exact error text here, not a description of it

No results We can’t find any text.

The exact error text, or the confidently wrong output. Say whether anything in the response warned you that something was wrong.

No results We can’t find any text.

Break 2. Take one of your chains and remove the middle stage. Describe what the chain now produces.

Paste the exact error text here, not a description of it

None, Stores in S3

What breaks, and whether it breaks obviously or quietly. Quiet degradation is the one worth writing about.

Breaks quietly. Stores unfiltered and unstructured medical data

Break 3. Recompute one Step 5 estimate at fifty times the volume.

Paste the exact error text here, not a description of it

NA

The new figure, whether it changes your service choice, and the volume at which your answer would flip.

$23000

$77000

8. Results

The numbers and outputs your own run produced.

Your eight row table

R1 | Amazon Textract: AnalyzeDocument (FORMS) | output-shape
R2 | Transcribe Medical → Comprehend Medical → S3 | compliance
R3 | Amazon Connect → Lex V2 → Lambda → Lex V2 | output-shape
R4 | Amazon Translate (batch) + custom terminology | cost
R5 | Rekognition: DetectModerationLabels → policy rules | capability-gap
R6 | Comprehend: DetectDominantLanguage → DetectPiiEntities | language
R7 | Comprehend: synchronous DetectSentiment → agent dashboard | latency
R8 | Textract (if scanned) → Bedrock sections → Bedrock brief | output-shape

Your chains, with each stage labeled

R2: Transcribe Medical (transcript) → Comprehend Medical (medication entities) → S3 (protected storage)
R6: DetectDominantLanguage (language code) → DetectPiiEntities (PII offsets/classes)
R3: Connect (call) → Lex V2 (intents/slots) → Lambda (booking) → Lex V2 (spoken response)
R8: Textract (text, if needed) → Bedrock (section summaries) → Bedrock (one-page brief)

Your Step 4 results, easy input and awkward input, with confidence scores if shown

Name: Dong Wellen - 81%
Name: Hu Sa Ka Nea – 90%

Your two cost estimates with sources and dates

$1850 – AWS Comprehend Pricing – 11/8/26
$1880 – AWS Rekognition Pricing - 11/8/26

9. Reflection

Your own words. No AI assistance on this part. Graders check these answers against the rest of your submission.

Question 1

Report your Step 2 prediction about which factor would dominate, then count what actually dominated in your finished table. Give both numbers out of eight and say what the gap taught you.

Compliance (1/8) was the predicted factor, but it was really output shape which dominated (3/8). I guess the output form is the most important thing to look for.

Question 2

Paste the awkward input you built in Step 4 and both results. Say whether the service told you it was struggling or returned something confidently wrong, and explain why that difference matters for a system you would put in front of customers.

Shane McQueen signature. It was confidently wrong. It matters because many forms are signed with cursive signatures

10. AI use disclosure

Required by the course AI Use Policy.

Tool used (write none if you used no AI):
ChatGPT (GPT-6)

If you used one: what you asked it, what you kept, and how you checked it

ChatGPT assisted in answering questions

11. Before you submit

- ☐ Every factor I used is one of the six allowed values, spelled exactly
- ☐ I quoted the constraint column, not my own opinion, in every why it fails line
- ☐ I actually ran a service in the console and captured a screenshot
- ☐ Both cost estimates show arithmetic, a URL, and a date
- ☐ I ran this entirely in the console and wrote no code
- ☐ Every prediction was written before I ran that step
- ☐ I recorded the actual result for every prediction, including the wrong ones
- ☐ I pasted the real error text for every break it item
- ☐ Work pushed to my GitHub portfolio repository and the URL is in here
- ☐ Reflection written by me with no AI assistance
- ☐ Files uploaded individually, no archive